MIL-P-13269A(Ord)
10 November 1960
SUPERSEDING
MIL-P-13269(Ord)
17 February 1954

MILITARY SPECIFICATION

PERISCOPE, TANK: M20 SERIES

1. SCOPE

1.1 This specification covers a series of periscopes used as sighting devices for armored vehicles and tanks. Each individual periscope has a designation affixed to the model number to identify minor design variations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-13462	Periscope M20 Series; Packaging of
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

FSC 1240

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DRAWINGS

Ordnance Corps

F7652791	Periscope M20
F7680994	Fixture, Coupling Snap
F8229219	Periscope M20A1
F8261628	Periscope M20A2
F8286740	Periscope M20A3
F8585211	Periscope M20A3C
F8286740	Periscope M20A3F
LSQAP-7652791	Supplementary Quality Assurance Provisions List for Periscope M20
LSQAP-8229219	Supplementary Quality Assurance Provisions List for Periscope M20A1
LSQAP-8261628	Supplementary Quality Assurance Provisions List for Periscope M20A2
LSQAP-8286740	Supplementary Quality Assurance Provisions List for Periscope M20A3 & M20A3F
LSQAP-8585211	Supplementary Quality Assurance Provisions List for Periscope M20A3C
IEL-7652791	Inspection Equipment List for Periscope M20
IEL-8229219	Inspection Equipment List for Periscope M20A1
IEL-8261628	Inspection Equipment List for Periscope M20A2
IEL-8286740	Inspection Equipment List for Periscopes M20A3 and M20A3F
IEL-8585211	Inspection Equipment List for Periscope M20A3C

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one periscope, representative of production, shall be made available to the Government to determine if the periscope meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

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3.3 Construction.- The periscope shall be manufactured in accordance with drawings applicable to the model periscope specified in the contract and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and optical characteristics to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specifications MIL-F-13926 and MIL-O-13830.

3.4.1 MIL-F-13926.

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.4.2 MIL-O-13830.

(a) Cleanliness
(b) Illuminated reticles

3.5 Environmental.

3.5.1 Storage temperatures.- The periscope shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all requirements of this specification at standard ambient temperatures (60 to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of minus 80 degrees Fahrenheit and plus 160 degrees Fahrenheit.

3.5.2 Operating temperatures.- The periscope shall meet the applicable requirements of 3.7.11 while exposed and thermally stabilized at ambient temperatures of minus 40 degrees Fahrenheit and plus 150 degrees Fahrenheit.

3.5.3 Vibration.- There shall be no visible damage or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds with the input coupling immobilized and the periscope mounted in its normal operating position. Subsequent to vibration, the periscope shall meet all of the following requirements of this specification.

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3.6 Head assembly.- With the mounting surface vertical (datum level B), the keyway horizontal (datum line A), and with a tangent to the two coupling pins in a vertical plane, the head assembly shall meet the following requirements:

3.6.1 Collimation.- With the incident line of sight vertical the following requirements shall be met:

3.6.1.1 The line of sight (reflection) shall be horizontal within 3 minutes.

3.6.1.2 The line of sight (reflection) shall be perpendicular to the mounting surface in lateral direction within 5 minutes.

3.6.1.3 The coupling when snapped against pin on Fixture F7680994 in both directions, shall not change the line of sight as specified in the collimation snap tests. This procedure shall be enacted without prior settling out rotation.

3.6.2 Plumb travel.- The line of sight shall not deviate from a plumb line by more than 0.25 mil when elevated or depressed 18 degrees coincident with the plumb line at the horizontal position.

3.6.3 Elevation accuracy.- The change in angle at the input coupling shall cause the same angular change in the line of sight within 0.25 mil. The coupling shall be set to allow a minimum of 22 degrees elevation and 18 degrees depression.

3.6.4 Definition.- The periscopes shall meet the definition requirements for the head assembly specified on the respective optical system drawings.

3.6.5 Backlash.- There shall be no backlash in excess of 0.1 mil when the mirror is rotated through travel, either in elevation or depression and returned to the original position.

3.7 Body assembly.- With mounting surface "A" horizontal and the boresight knobs set to "3" the body assembly shall meet the following requirements.

3.7.1 Collimation.- The line of sight through the boresight mark at the intersection of the cross shall be parallel with the mounting surface "A" (F7652792) within one mil, and perpendicular to datum line "B" in a lateral direction within one mil, including locking error.

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3.7.1.1 There shall be no deviation in the line of sight in excess of 0.05 mil as a result of actuating the boresight locking lever.

3.7.2 Backlash.

3.7.2.1 Boresight knobs.- Backlash in the boresight knobs shall not exceed 0.1 mil.

3.7.2.2 Diopter movement.- Backlash in the diopter movement shall not exceed 1/8 diopter.

3.7.3 Image tilt.- The image of a horizontal target shall be horizontal within one degree as measured by an auxiliary telescope adjacent to the eyepiece. One degree is equivalent to the angle subtended by 2.6 mils at the edge of the field when the tilt is measured over the full field of view.

3.7.4 Parallelism of image and reticle.- The horizontal reticle line shall not be out of parallelism with the image of a horizontal target reference line by more than the angle subtended by a 0.2 mil at the extremity of the horizontal reticle line when the other extremity of this line is in coincidence with the target reference line.

3.7.5 Parallax.- Parallax at the optical axis shall not exceed 0.1 mil when viewing a target at 500 yards to infinity.

3.7.6 Eyepiece focus.- The diopter scale shall indicate zero within plus or minus 0.25 diopter, including backlash when the reticle is brought into sharp focus. From the zero position the diopter scale shall have at least three diopters movement in each direction as read on the eyepiece scale.

3.7.7 Reticle lock.- The locked reticle shall not have shifted more than 0.2 mil subsequent to vibration.

3.7.8 Reticle shift.- Any movement of the horizontal and vertical adjusting reticle knobs shall not cause the reticle to shift more than 0.2 mil in a direction perpendicular to the movement of the reticle cell.

3.7.9 Reticle illumination.- With the reticle illuminated by Instrument Light, M36 or equivalent, the reticle marking shall appear clearly defined when observed in a dark room.

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3.7.10 Definition.- Resolving power on the optical axis for a parallel bar test object shall be 6.5 seconds of arc or better.

3.7.11 Torque.

3.7.11.1 Boresight knobs.- The running torque required to rotate each boresight knob in both directions, when disengaged, shall not exceed 5 pound inches at standard ambient temperatures (60 degrees to 90 degrees Fahrenheit) and the torque shall not change from the value obtained at standard ambient temperatures by more than 2 pound inches when subjected to the temperatures specified in 3.5.2.

3.7.11.2 Diopter adjustment.- The running torque required to rotate the diopter adjustment shall not exceed seven pound inches at standard ambient temperatures (60 degrees to 90 degrees Fahrenheit) and the torque shall not change from the value obtained at standard ambient temperatures by more than three pound inches when subjected to the temperatures specified in 3.5.2.

3.8 Identification marking.- Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

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(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.4 Production model.- Unless otherwise specified in the contract, one periscope shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

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4.4.1 Production model failure.- Should the production model fail to pass the prescribed examinations and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

4.5.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.2 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)- 7652791, 8229219, 8261628, 8286740 and 8585211.

4.5.2.1 Contents of SQAP.- The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications, and to reject individual non-conforming items.

4.5.2.2 Contractor's inspection for all components not covered by SQAP.- Inspection for these components shall be in accordance with the inspection procedures of the written inspection plan required by 4.2 and 4.2.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the periscopes.

4.5.2.3 Disposition of non-conforming product.- Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

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4.7 Acceptance tests.

4.7.1 Classification of defects.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Boresight knob torque	3.7.11.1	4.9.3.11.1	Device, Special testing (DST)
102. Diopter adjustment torque	3.7.11.2	4.9.3.11.2	DST

MINOR: NONE DEFINED

NOTE: The above tests shall be performed while the periscopes are thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
101. Physical failure post vibration	3.5.3	4.9.1.4	Visual-Tactile
102. Cleanliness	3.4.2	4.9.1.5	Visual
Head Assembly			
103. Collimation	3.6.1	4.9.2.1 4.9.2.1.1	Device, Special Testing (DST)
104. Plumb travel	3.6.2	4.9.2.2	DST
105. Elevation accuracy	3.6.3	4.9.2.3	DST
106. Definition	3.6.4	4.9.2.4	DST
107. Backlash	3.6.5	4.9.2.5	DST
Body Assembly			
108. Collimation	3.7.1	4.9.3.1	DST
109. Boresight knob backlash	3.7.2.1	4.9.3.2.1	DST
110. Diopter movement backlash	3.7.2.2	4.9.3.2.2	DST
111. Image tilt	3.7.3	4.9.3.3	DST
112. Parallelism of image and reticle	3.7.4	4.9.3.4	DST
113. Parallax	3.7.5	4.9.3.5	DST
114. Eyepiece focus	3.7.6	4.9.3.6	DST
115. Reticle lock	3.7.7	4.9.3.7	DST
116. Reticle shift	3.7.8	4.9.3.8	DST
117. Reticle illumination	3.7.9	4.9.3.9	Visual
118. Definition	3.7.10	4.9.3.10	DST
119. Boresight knob toruqe	3.7.11.1	4.9.3.11.1	DST
120. Diopter adjustment torque	3.7.11.2	4.9.3.11.2	DST
121. Identification marking	3.8	4.9.4	Visual

MINOR: NONE DEFINED

NOTE: The above tests shall be performed at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

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4.8 Inspection equipment.

4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for reveiw by the Government. The government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.8.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance except in those cases specified otherwise. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.2 Government provisions.- Test equipment listed on Inspection Equipment List (IEL-7652791) for Periscope M20, (IEL-8229219) for Periscope M20A1, (IEL-8261628) for Periscope M20A2, (IEL-8585211) for Periscope M20A3C and (IEL-8286740) for Periscopes M20A3 and M20A3F, when available, will be used for Government product inspection (reference 4.3(b)), and to settle disputes (reference 4.2.1). Drawings of this equipment may be furnished the contractor for informational purposes.

4.8.3 Accuracy requirements for special testing devices.

4.8.3.1 Device to test storage and operating temperatures.- The device shall be capable of meeting the specified temperature requirements within the test equipment tolerances specified in Military Specification MIL-F-13926.

4.8.3.2 Device to test vibration.- The device shall be capable of accuracies as good or better than those obtainable with the device shown on Drawings F7560085, F7560503 and F7560583.

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4.8.3.3 Device to test head assembly collimation, plumb travel, elevation accuracy, definition and backlash.- A mandatory testing device, which shall be in strict accordance with all requirements and specifications contained in and pertinent to Ordnance Drawings F7560570 (Fixture), D7560514 (Set-up and Operating Instructions), F7560516 (Gage Setting), F7560517 (Eyepiece 6-power), F7560518 (Light Assembly Collimator) and F7680994 (Fixture and Set-up and Operating Instructions).

4.8.3.4 Device to test body assembly collimation, backlash, boresight knobs, diopter movement, image tilt, parallelism of image and reticle, parallax, eyepiece focus, reticle shift, reticle illumination, and definition.- A mandatory testing device, which shall be in strict accordance with all requirements and specifications contained in and pertinent to Ordnance Drawings D7560594 (Fixture), D7560541 (Set-up and Operating Instructions), D7560565 (Gage Setting) F7560516 (Gage Setting), F7560517 (Eyepiece 6-power), F7560518 (Light Assembly Collimator), F7680994 (Fixture and Set-up and Operating Instruction). The following listed equipment shall be used in addition to the above listed equipment, for testing Periscope Body Assemblies M20A3C and M20A3F, Ordnance Drawing F8617986 (Fixture), Ordnance Drawing B8618032, (Set-up Instructions), Ordnance Drawing B8618031 (Operating Instructions) and Ordnance Drawing D8618013 (Adapter).

4.8.3.5 Device to test torque of boresight knobs, diopter adjustment.- This device shall be capable of accuracies as good or better than those obtainable with the devices shown on Drawing F7560571 and F7560506.

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Storage temperatures.- This test shall be conducted in accordance with Procedure II of Specification MIL-F-13926 to the condition specified. Upon return to the standard ambient temperature, the periscope shall be inspected for damage and then subjected to the tests specified in 4.9.1.3 to 4.9.3.11.2 inclusive, and shall meet the requirements specified. This test shall be conducted on the production model only.

4.9.1.2 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" and not raised beyond condition "G". When the periscope is thermally stabilized at condition "C" and "G", it shall be subjected to the tests specified in 4.9.3.11, and shall meet the requirements specified.

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4.9.1.3 Vibration.- Each periscope in the sample shall be mounted on a special testing device and vibrated in accordance with the requirements as specified.

4.9.1.4 Physical failure, post vibration.- Subsequent to vibration, the periscope shall be subjected to a visual and tactile examination for damage such as: loose, bent or broken parts, glass chips, etc. The periscope shall then be subjected to the tests specified in 4.9.1.2 to 4.9.4 inclusive.

4.9.1.5 Cleanliness, post vibration.- The optical elements of the periscope shall be visually examined for cleanliness in accordance with Specification MIL-O-13830. The internal surfaces shall show no evidence of moisture, or dust particles.

4.9.2 Head assembly.- The head assembly to be tested shall be mounted on a special testing device with the mounting surface vertical and keyway horizontal as specified.

4.9.2.1 Collimation.- With the incident line of sight vertical, the following tests shall be made.

4.9.2.1.1 The head assembly shall be firmly clamped against the mounting surface of the special testing device. When observed through the telescope, the image of the dot in the center of the second reticle must be entirely within the large rectangle of the first reticle. The line of sight (reflection) will then be within the horizontal and perpendicular tolerance as specified.

4.9.2.1.2 The coupling on head assemblies for Periscopes M20 and M20A2 shall be rotated in a clockwise direction to its maximum spring tension and released suddenly, allowing coupling face to snap against pin. Repeat this operation for at least six successive times; the maximum deflection of the line of sight from the established value shall be noted. The coupling face which has just been snapped shall be held securely against the pin and a torque of 6 to 8 pound-inches shall be imposed in a clockwise direction, then the coupling shall be released suddenly allowing the opposite face of the coupling to snap against the opposite side of the pin. After this operation and each successive operation for at least six times, the line of sight shall not vary from the maximum deflection, established above, by more than 0.5 mil, but the summation of the maximum change on each side of the established value shall not exceed 0.5 mil.

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4.9.2.1.3 On head assemblies of Periscope M20A1, M20A3, M20A3C and M20A3F, the same procedure as specified in 4.9.2.1.2 shall be used except that the line of sight shall not change from maximum deflection value by more than 0.2 mil, but the summation of the maximum change on each side of the established value shall not exceed 0.2 mil.

4.9.2.2 Plumb travel.- Using the special testing device, observe the luminous dot from the collimator through the eyepiece of the telescope. Adjust the prism by means of the knurled screws until the dot has been centered in the circle of the reticle. Move the lever handle to the required angular displacement. The dot shall remain inside the circle of the reticle to comply with the specified requirements.

4.9.2.3 Elevation accuracy.- This test can be performed in conjunction with 4.9.2.2 above, except that the input coupling shall be set to the required angular movement. The dot shall remain inside the circle of the reticle to comply with the specified requirements.

4.9.2.4 Definition.- Definition shall be tested on the special testing device. The resolution pattern on the reticle is designated in seconds of arc. Adjust the telescope for the sharpest focus of the resolution reticle pattern. The test pattern must resolve simultaneously in each of the four meridians without adjusting the diopter setting on the telescope more than the specified requirements.

4.9.2.5 Backlash.- This test can be performed in conjunction with the elevation accuracy test in 4.9.2.3. Move the lever handle through full travel, either in depression or elevation, and return to the original position. The backlash of the mirror shall not exceed the specified requirements.

4.9.3 Body assembly.- The body assembly to be tested shall be mounted on a special testing device with the mounting surface horizontal and the boresight knobs set to "3", then the following tests shall be performed.

4.9.3.1 Collimation.- View the reticle of the periscope body through the collimator eyepiece. The horizontal and vertical cross-lines of the periscope reticle must be within, or in coincidence with, the small square of the collimator reticle to meet the specified requirements.

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4.9.3.1.1 To test for deviation in the line of sight, actuate the boresight locking lever and repeat 4.9.3.1, above. There shall be no deviation in excess of the specified requirements.

4.9.3.2 Backlash.

4.9.3.2.1 Boresight knobs.- With the backlash excluded for direction of rotation, record position of knob and line of sight. Rotate knob one-half revolution, then with backlash included, return the knob to the recorded position without overtravel and record the position of the line of sight. The difference between the two recorded positions of the line of sight is the amount of backlash. Backlash in each boresight knob shall not exceed the tolerance specified.

4.9.3.2.2 Diopter movement.- This test shall be conducted concurrent with the eyepiece focus test in 4.9.3.6. With backlash excluded for direction of travel, rotate the diopter scale of the body assembly from the zero position to plus or minus 2 diopters, then with backlash included, return the diopter scale to the zero position. Any adjustment of the observing telescope to obtain best focus of the reticle image shall not exceed the tolerance specified.

4.9.3.3 Image tilt.- Rotate the wedge until the upper vertical line of the collimator reticle is in coincidence with the plumb cross of the telescope reticle. When this has been done, the lower vertical line must be within the limit marks of the observing telescope reticle to meet the specified requirements.

4.9.3.4 Parallelism of image and reticle.- This test can be performed in conjunction with the collimation test in 4.9.3.1. View the periscope reticle through the collimator eyepiece to check the reticle and image parallelism. The extreme ends of the center of the horizontal line of the periscope reticle must be inside of the 0.2 mil lines to meet the specified requirements.

4.9.3.5 Parallax.- Parallax of the body optical system shall be checked by using the 0.2 mil reference lines of the collimator reticle as a unit of measure with which to estimate parallax. Parallax will be recognized as any apparent movement between the two images when the observer's head is moved from side to side and up and down. Parallax shall not exceed the specified tolerance.

4.9.3.6 Eyepiece focus.- Set the focusing ring of the observing telescope on zero and bring the dot of the reticle into sharpest focus by means of the periscope eyepiece diopter scale. At this point of sharpest focus, the diopter scale shall indicate zero within the specified tolerance. From the zero position, the diopter scale shall be rotated to insure at least three diopters movement in each direction.

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4.9.3.7 Reticle lock.- Select a point on the horizontal reticle line of the periscope which is in coincidence with the horizontal center of the collimator reticle. The vertical reticle line of the periscope shall be within the square of the collimator reticle. Lock both boresight knobs and perform the vibration test in 4.9.1.3. After vibration, the selected point on the horizontal line shall be within the specified tolerance line and shall be within the square of the collimator reticle.

4.9.3.8 Reticle shift.- With the horizontal and vertical cross-lines of the periscope reticle in the exact center collimation position both boresighting knobs should be at position "3" on the boresight scales. Rotate the boresight knobs; first to one extreme position, then the other. Reticle shift shall be within the specified tolerance.

4.9.3.9 Reticle illumination.- Illumination of the reticle is checked visually using Instrument Light M36 or an equivalent source. The reticle markings shall appear clearly defined when examined in a darkened area.

4.9.3.10 Definition.- The observing telescope shall be placed into position. Observe the collimator reticle through the observing telescope eyepiece. The number to the side of each resolution pattern represents the actual seconds of arc resolution. The resolving power shall be no less than the specified requirements.

4.9.3.11 Torque.

4.9.3.11.1 Boresight knobs.- Using a special testing device, lock the device around the boresight knob and rotate the unlocked knob by the device in both directions. The torque required to rotate the knob shall not exceed the value specified for the temperature indicated. This test shall be performed on both boresight knobs.

4.9.3.11.2 Diopter adjustment.- Repeat the above test in 4.9.3.11.1 on the diopter scale. The torque required to rotate the diopter scale in both directions shall not exceed the value specified for the temperature indicated.

4.9.4 Identification marking.- Markings shall be visually inspected for conformance to Standard MIL-STD-130.

4.9.5 Quality assurance provisions for packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-13462.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-13462. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Ordering data.- Procurement document should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.2 Definitions.

6.2.1 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.2.2 Tactile.- Skillful manipulation or sensing.

6.3 Periscope M20A3F.- The values on the boresight knob scales of the Periscope M20A3F are calibrated for ranges in meters.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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